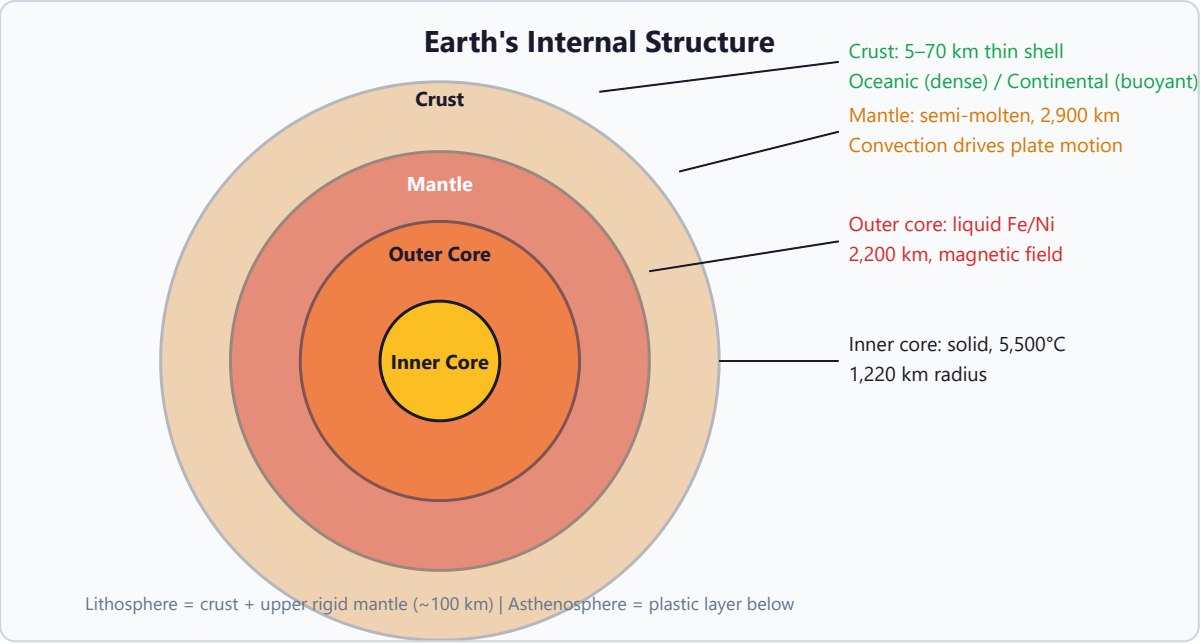
Topic 1: Tectonic Processes & Hazards

1.1 The Structure of the Earth

Plate tectonics: The theory that the Earth's lithosphere is divided into ~15 major rigid plates (and several minor ones) that move relative to each other, driven by convection currents in the mantle, slab pull at subduction zones, and ridge push at mid-ocean ridges. First comprehensively articulated in the 1960s, unifying Wegener's continental drift hypothesis (1912) with Hess's sea-floor spreading theory (1962).

Earth's Internal Structure

- **Inner core** — solid iron and nickel, ~5,500°C, radius ~1,220 km. Solid despite extreme temperature because of immense pressure (~360 GPa).
- **Outer core** — liquid iron and nickel, ~2,200 km thick. Convection in the liquid outer core generates Earth's magnetic field (geodynamo).
- **Mantle** — semi-molten silicate rock, ~2,900 km thick. The upper mantle contains two key layers: the rigid **lithosphere** (crust + uppermost mantle, ~100–250 km thick, broken into tectonic plates) and the plastic **asthenosphere** (partially molten, allowing plates to move over it).
- **Crust** — the thin outer shell. *Continental crust*: thick (25–70 km), less dense (~2.7 g/cm^3), granitic composition, geologically old (up to 4 billion years), cannot be subducted. *Oceanic crust*: thin (5–10 km), denser (~3.0 g/cm^3), basaltic composition, geologically young (<200 million years, continuously recycled at subduction zones).



Evidence for Plate Tectonics

- **Continental fit** — jigsaw-like fit of South America and Africa (first noted by Wegener, 1912). Computer modelling shows a fit at the continental shelf edge at 500 fathoms depth.
- **Geological evidence** — matching rock types, mountain chains (e.g. Caledonian Mountains of Scotland/Scandinavia match Appalachians of eastern USA), and fossil distributions across continents now separated by oceans. *Glossopteris* fern fossils found on South America, Africa, India, Australia, and Antarctica — impossible if continents were always in current positions. *Mesosaurus* freshwater reptile fossils found in both Brazil and South Africa.
- **Palaeoclimatic evidence** — glacial deposits (tillites) of the same age found in India, Africa, South America, and Australia, suggesting they were once joined near the South Pole (Gondwanaland). Coal deposits in Antarctica indicate it was once in a tropical/subtropical position.
- **Palaeomagnetism** — symmetrical patterns of magnetic striping on the ocean floor either side of mid-ocean ridges. As new basaltic crust forms at ridges, iron minerals align with Earth's magnetic field. Periodic magnetic field reversals create a "barcode" pattern of normal and reversed polarity stripes, proving new crust is continuously created and spreading.
- **Sea-floor spreading** — younger rocks near mid-ocean ridges, progressively older rocks further away (Harry Hess, 1960s). Deep-sea drilling confirmed this: the oldest oceanic crust is ~200 million years old (at subduction zones), compared to continental crust up to 4 billion years old.
- **GPS measurements** — modern satellite technology directly measures plate movements: North Atlantic widening at ~2.5 cm/yr; Pacific plate moving NW at ~7 cm/yr; India moving N into Eurasia at ~5 cm/yr.

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Provides irrefutable proof of plate motion.

- **Distribution of earthquakes and volcanoes** — concentrated along plate boundaries, clearly delineating plate edges. Deep-focus earthquakes at subduction zones trace the descending slab (Benioff Zone).

Driving Mechanisms of Plate Movement

- **Mantle convection** — heat from radioactive decay in the core and mantle creates convection currents. Hot material rises at constructive boundaries (mid-ocean ridges), spreads laterally, cools, and sinks at destructive boundaries (subduction zones). This drag moves plates, but is now considered less important than slab pull.
- **Slab pull** — at subduction zones, cold, dense oceanic lithosphere sinks into the mantle under gravity. The weight of the descending slab "pulls" the rest of the plate behind it. Now considered the **dominant driving mechanism**, responsible for ~90% of plate motion. Explains why plates with large subduction zones (e.g. Pacific plate) move fastest.
- **Ridge push** — newly formed crust at mid-ocean ridges sits higher (because it is hotter and less dense) than older, cooler crust. Gravity causes the elevated ridge to push the plate away from the ridge axis. Less important than slab pull but contributes to motion.
- **Mantle plumes/hot spots** — localised upwellings of abnormally hot mantle material create volcanic activity away from plate boundaries (e.g. Hawaii, Yellowstone, Iceland). As the plate moves over the stationary plume, it creates a chain of progressively older volcanic islands (Hawaiian chain).

1.2 Plate Boundaries

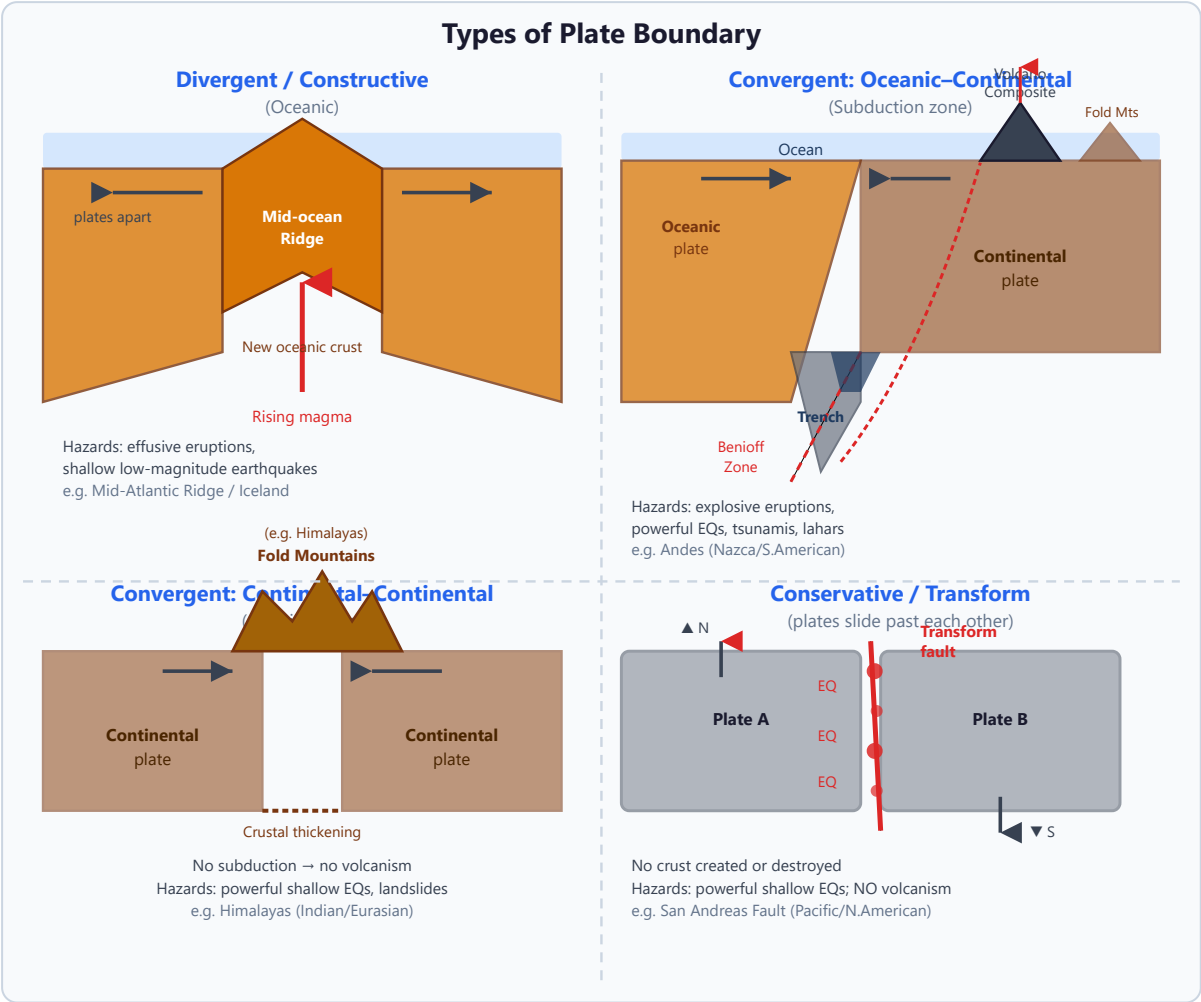
Types of Plate Boundary

Boundary Type	Movement	Key Features	Hazards	Examples
Divergent (constructive)	Plates move apart	Mid-ocean ridges (oceanic): new crust formed from rising magma. Rift valleys (continental): crust	Volcanic eruptions (basaltic, effusive, low risk); shallow earthquakes (generally <M 6). Relatively low hazard risk.	Mid-Atlantic Ridge (Iceland); East African Rift Valley (Kenya, Ethiopia, Tanzania)

		stretches, thins, and fractures.		
Convergent: oceanic-continental	Plates move together; oceanic subducts	Ocean trench, subduction zone (Benioff Zone), volcanic arc (composite volcanoes on continental plate), fold mountains	Powerful earthquakes (deep and shallow); explosive volcanic eruptions (andesitic/rhyolitic); tsunamis; lahars	Andes (Nazca under South American); Cascades (Juan de Fuca under North American)
Convergent: oceanic-oceanic	Plates move together; denser plate subducts	Ocean trench, volcanic island arc	Powerful earthquakes; explosive volcanic eruptions; tsunamis	Mariana Trench (Pacific under Philippine); Japan; Caribbean island arc
Convergent: continental-continental (collision)	Plates collide; neither subducts	Massive fold mountains; crustal thickening and uplift; no volcanism (no subduction = no melting)	Powerful shallow earthquakes; landslides	Himalayas (Indian-Eurasian); Alps (African-Eurasian)
Conservative (transform)	Plates slide past each other	Transform fault; no crust created or destroyed; plates move in same or opposite directions at different rates	Shallow, powerful earthquakes (sudden release of built-up friction); NO volcanism	San Andreas Fault (Pacific-North American); North Anatolian Fault (Turkey); Alpine Fault (New Zealand)

Exam Tip: Edexcel expects you to understand why hazards vary at different plate boundaries. Divergent boundaries produce low-viscosity basaltic magma (effusive eruptions, lower risk). Convergent boundaries produce high-viscosity andesitic/rhyolitic magma (explosive eruptions, higher risk) because subducting oceanic crust and seawater trigger partial melting with high silica

content. Conservative boundaries produce no magma (no volcanic hazard) but store enormous strain energy released as powerful earthquakes.



1.3 Volcanic Hazards

Volcano: An opening in the Earth's crust through which magma, ash, and gases are ejected. The nature of an eruption depends primarily on magma viscosity, which is controlled by silica content, temperature, and gas content. ~1,500 potentially active volcanoes exist; ~50–70 erupt each year.

Magma Types and Eruption Styles

Magma Type	Silica Content	Viscosity	Temperature	Gas Content	Eruption Style	Volcano Type

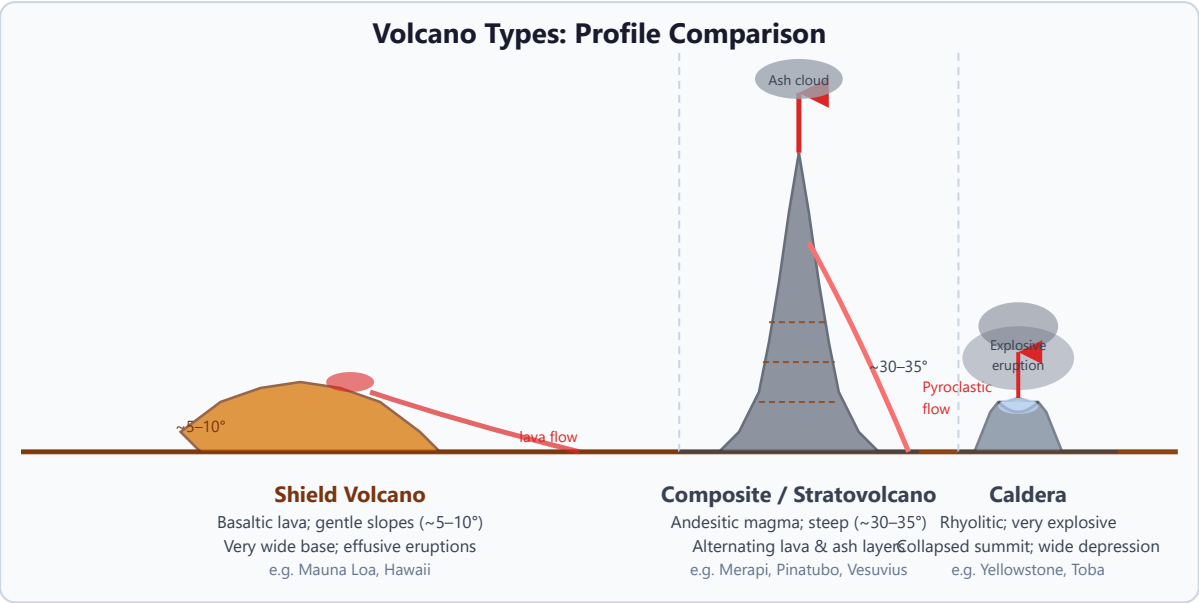
Basaltic	Low (~50%)	Low (runny)	1,000–1,200°C	Low	Effusive: lava flows, fountains	Shield (e.g. Mauna Loa)
Andesitic	Medium (~60%)	Medium	800–1,000°C	Medium	Mixed: explosive and effusive	Composite/stratovolcano
Rhyolitic	High (~70%)	Very high (sticky)	650–800°C	High (trapped)	Explosive: pyroclastic flows, ash clouds	Caldera/supervolcano

Volcanic Hazards

- **Lava flows** — basaltic flows travel 10–60 km/h, covering large areas. Slow enough to evacuate from but destroy everything in their path. Buildings, roads, farmland inundated. Kilauea, Hawaii (2018) destroyed 700 homes.
- **Pyroclastic flows** — superheated avalanches of gas, ash, and rock fragments at 100–700 km/h and 200–1,000°C. The most lethal volcanic hazard. Impossible to outrun. Mont Pelee, Martinique (1902): pyroclastic flow killed 29,000 in Saint-Pierre in minutes.
- **Lahars** — volcanic mudflows formed when ash mixes with water (rain, snowmelt, crater lake). Travel 20–100 km/h, following river valleys, reaching 100+ km from the volcano. Nevado del Ruiz (1985): lahars killed 23,000 in Armero, Colombia despite 2 hours' warning.
- **Tephra and ash fall** — rock fragments of all sizes ejected into atmosphere. Fine ash travels hundreds/thousands of km. Collapses roofs (>15 cm accumulation), contaminates water, damages crops, disrupts aviation (Eyjafjallajokull 2010: 100,000 flights cancelled, $1.7 billion airline losses), respiratory hazards.
- **Volcanic gases** — CO_2 (heavier than air, pools in depressions: Lake Nyos, Cameroon 1986 killed 1,700 from CO_2 suffocation), SO_2 (acid rain, respiratory damage, climate cooling via stratospheric sulphate aerosols: Pinatubo 1991 cooled Earth ~0.5°C for 2 years), H_2S, HF.
- **Tsunamis** — generated by volcanic island collapse, submarine eruptions, or caldera formation. Krakatoa (1883): tsunamis up to 40 m killed 36,000. Hunga Tonga-Hunga Ha'apai (2022): submarine

eruption generated tsunamis across the Pacific.

- **Jokullhlaups** — glacial outburst floods caused by subglacial volcanic eruptions melting ice. Specific to Iceland and other glaciated volcanic regions.



Volcanic Explosivity Index (VEI)

A logarithmic scale from 0 (gentle effusive) to 8 (mega-colossal) measuring eruption magnitude based on volume of material ejected and plume height. Each level represents a 10x increase. VEI 5+ eruptions are relatively rare: Pinatubo (1991) was VEI 6; the most recent VEI 7 was Tambora (1815, which caused the "Year Without a Summer" in 1816). VEI 8 (supervolcano): last was Toba ~74,000 years ago; Yellowstone has potential but low probability.

Case Study: Mount Merapi, Indonesia (2010) — Developing Country Volcanic Hazard

Mount Merapi is one of the world's most active stratovolcanoes, located in central Java on a destructive boundary (Indo-Australian plate subducting beneath the Eurasian plate). 25 million people live within 30 km — attracted by extremely fertile volcanic soils. The October–November 2010 eruption (VEI 4) produced pyroclastic flows reaching 15+ km from summit, ash columns to 18 km, and lahars during monsoon season. **Impacts:** 353 deaths, 350,000 displaced, 2,682 houses destroyed, ~$600 million economic damage, livestock killed, agriculture devastated. **Response:** PVMBG (volcanology agency) raised alert to maximum level 4 days before main eruption; 20,000 evacuated. Government established 20 km exclusion zone. However, many residents returned prematurely to tend livestock, increasing casualties. Post-eruption: improved real-time GPS and seismic monitoring network, updated hazard maps with 3D pyroclastic flow simulations, permanent evacuation routes

and concrete shelters constructed. **Wider significance:** Merapi illustrates the challenge of managing volcanic hazards in densely populated developing countries where livelihoods depend on volcanic resources (fertile soil, geothermal energy, tourism).

Case Study: Eyjafjallajokull, Iceland (2010) — Global Impacts of a Moderate Eruption
Iceland sits on both the Mid-Atlantic Ridge (divergent boundary) and a mantle hot spot, making it one of the most volcanically active places on Earth. Eyjafjallajokull erupted March–May 2010 (VEI 4). The eruption was moderate in volcanic terms but its impacts were globally significant because: (1) phreatomagmatic interaction (magma-glacial meltwater contact) fragmented ash into exceptionally fine particles that stayed airborne; (2) prevailing jet stream carried ash directly across European airspace; (3) risk-averse aviation policy imposed blanket flight bans. **Impacts:** ~100,000 flights cancelled over 6 days, 10 million passengers stranded, airline losses ~$1.7 billion, global supply chain disruption (Kenyan flower exports fell 50%, BMW halted production due to missing parts), but zero direct deaths from eruption. Iceland locally: 800 evacuated from jokulhlaup flooding, farmland covered in fluoride-rich ash harming livestock. **Lessons:** demonstrated globalised interconnections (a small eruption in Iceland disrupted the global economy), led to revised aviation volcanic ash protocols (from blanket bans to defined ash concentration thresholds), improved volcanic ash dispersion modelling.

1.4 Seismic Hazards

Earthquake: A sudden release of accumulated strain energy in the Earth's lithosphere, generating seismic waves that cause ground shaking. Most earthquakes occur at plate boundaries where friction between moving plates causes stress to build, then release suddenly when friction is overcome. The point of energy release is the **focus (hypocentre)**; the point directly above on the surface is the **epicentre**.

Types of Seismic Wave

- **Primary (P) waves** — fastest (6–8 km/s in crust), compressional, travel through solids and liquids, cause back-and-forth shaking. Arrive first at seismographs.
- **Secondary (S) waves** — slower (3–5 km/s), shear waves, travel only through solids, cause side-to-side and up-and-down motion. More destructive than P waves.

- **Surface waves (Love and Rayleigh)** — slowest but most destructive, travel along the surface causing rolling and shearing motion. Responsible for most earthquake damage.

Measuring Earthquakes

Scale	Measures	Key Features
Moment Magnitude Scale (M_w)	Total energy released (seismic moment = rigidity × fault area × displacement)	Logarithmic: each whole number = ~32x more energy. No upper limit. Used universally by seismologists. Replaced the Richter scale.
Modified Mercalli Scale (MMI)	Intensity of shaking and observed damage	Qualitative (I = not felt, XII = total destruction). Varies with distance from epicentre, local geology, building quality. Useful for comparing impacts at different locations.

Earthquake Hazards

- **Ground shaking** — the primary hazard. Intensity depends on magnitude, distance from epicentre, depth of focus, local geology (soft sediment amplifies shaking), and building quality.
- **Ground displacement/rupture** — permanent displacement along the fault line. Can sever roads, pipelines, railways, and buildings straddling the fault.
- **Liquefaction** — saturated, loose sediment (sand, silt) loses strength during shaking and behaves like liquid. Buildings sink, tilt, or collapse. Christchurch 2011: extensive liquefaction in suburbs built on river silt; 10,000 homes damaged beyond repair.
- **Landslides and rockfalls** — shaking destabilises slopes. Nepal 2015: thousands of landslides blocked roads and buried villages. Kashmir 2005: landslides killed thousands in mountain valleys.
- **Tsunamis** — generated by submarine earthquakes that vertically displace the sea floor (typically M 7+, shallow focus, at subduction zones). Waves travel at 500–800 km/h across open ocean (barely noticeable), but slow and build to 10–40 m in shallow coastal water.
- **Fires** — ruptured gas mains, broken electrical cables, and overturned stoves start fires. Fire services often cannot respond due to damaged roads and water mains. Kobe 1995: 7,000 buildings destroyed by fire.
- **Secondary impacts** — disease outbreaks (contaminated water, overcrowded shelters), economic disruption, displacement, psychological trauma, infrastructure destruction.

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Factors Affecting Earthquake Impact

Factor	Effect on Impact
Magnitude	Higher magnitude = more energy released = wider area affected = more damage potential
Depth of focus	Shallow (<70 km) generally more destructive; energy concentrated near surface. Deep (>300 km) less damage at surface.
Distance from epicentre	Shaking intensity decreases with distance (attenuates). But effects can be felt 100s of km away for large events.
Population density	Urban earthquakes affect more people. Rural earthquakes may cause few casualties despite high magnitude.
Building quality	Earthquake-resistant design (steel frame, base isolation, cross-bracing, ductile materials) dramatically reduces collapse. Unreinforced masonry, adobe, and concrete frame buildings are most vulnerable.
Ground conditions	Soft sediment/landfill amplifies shaking (2–5x) and risks liquefaction. Solid bedrock transmits less damaging shaking.
Time of day	Nighttime earthquakes cause more casualties (people sleeping in buildings). Daytime in school/work buildings can also be devastating if poorly constructed.
Level of development	HICs: better building codes, emergency services, insurance, infrastructure resilience. LICs: weaker governance, informal construction, limited emergency capacity.
Preparedness	Education, drills, early warning systems, emergency plans, stockpiled supplies all reduce casualties.
Secondary hazards	Tsunamis, landslides, fires, and dam failures can multiply the death toll far beyond the earthquake itself.

Case Study: Haiti Earthquake (12 January 2010) — LIC

A magnitude 7.0 earthquake struck 25 km west of Port-au-Prince at a shallow depth of 13 km on the

Enriquillo-Plantain Garden transform fault. **Impacts:** ~220,000–316,000 deaths (estimates vary), 300,000 injured, 1.5 million displaced, 250,000 homes destroyed, 30,000 commercial buildings collapsed including the Presidential Palace, UN headquarters, and hospitals. Cost: $8 billion (>120% of GDP). **Why so devastating:** extreme poverty (GDP per capita $660, poorest country in Western Hemisphere), virtually no building codes (unreinforced concrete frame construction), dense population (2.5 million in Port-au-Prince), weak governance and institutional capacity (the government itself was crippled — parliament, ministries destroyed), no earthquake preparedness programme, poor infrastructure (roads blocked by rubble, port damaged, airport limited capacity). **Response:** international aid mobilised ($13.5 billion pledged) but delivery was chaotic — Port-au-Prince's port was damaged, airport could handle only limited flights, roads were blocked. Cholera outbreak (introduced by UN peacekeepers) killed 10,000+ in following years. **Recovery:** painfully slow. 5 years later, 85,000 people still in temporary shelters. Reconstruction hampered by unclear land ownership, corruption, and ongoing political instability. Haiti demonstrates how vulnerability (poverty, governance, building quality) can be more important than magnitude in determining disaster severity.

Case Study: Tohoku, Japan (11 March 2011) — HIC

A magnitude 9.0 megathrust earthquake offshore NE Japan, focus at 32 km depth, on the destructive boundary where the Pacific plate subducts beneath the North American plate. The 4th largest earthquake ever recorded. **Earthquake impacts:** ground shaking lasted 6 minutes (exceptionally long), significant infrastructure damage despite Japan's world-leading earthquake engineering (bridges, roads, railways damaged). **Tsunami:** waves up to 40 m struck the coast within 30 minutes, overwhelming sea walls designed for 5–10 m waves. Penetrated up to 10 km inland. **Nuclear disaster:** Fukushima Daiichi power plant lost cooling systems when the 14 m tsunami overtopped its 5.7 m sea wall and flooded backup diesel generators. Three reactor meltdowns followed — world's worst nuclear disaster since Chernobyl. 154,000+ evacuated from 20 km exclusion zone; many areas remain uninhabitable. **Total impacts:** 19,747 deaths (93% from drowning in tsunami), 6,242 injured, 470,000 evacuated, 127,000 buildings destroyed. Economic cost: ~$235 billion (costliest natural disaster in history). **Response:** Japan's early warning system provided 8–30 minutes tsunami warning; many evacuated but many did not reach higher ground in time. Self-Defence Forces (100,000 troops) deployed within hours. **Lessons:** even the world's best-prepared country can be overwhelmed by extreme events. Japan has since built 12.5 m sea walls, revised tsunami warning systems, enhanced nuclear safety regulations, and created memorial parks with evacuation towers. The disaster also triggered Germany's decision to phase out nuclear power entirely by 2023.

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1.5 Hazard Management and Governance

Approaches to Hazard Management

Approach	Description	Examples	Effectiveness
Prediction and monitoring	Using scientific instruments and data to forecast when and where hazards will occur	Seismometers, tiltmeters, GPS ground deformation, gas spectrometers, satellite thermal imaging, historical recurrence intervals	Volcanoes: moderately predictable (precursors: seismicity, gas, deformation). Earthquakes: NOT reliably predictable — no proven method exists. Tsunamis: excellent detection once generated (DART buoys, seismic data), but limited warning time for near-field events.
Protection (engineering)	Physical measures to reduce damage from hazards	Earthquake-resistant buildings (base isolation, cross-bracing, dampers), sea walls, tsunami barriers, lava diversion channels, fire-resistant construction	Highly effective for earthquakes in well-enforced building code environments (Japan, Chile). Sea walls: effective up to design limits but can be overtopped (Tohoku 2011). Costly — not affordable for many LICs.
Preparedness (planning)	Planning and education to enable effective response	Emergency plans, evacuation routes, drills (Japan: "Disaster Prevention Day" every 1 September), stockpiled supplies, community training, hazard mapping, public education campaigns	Very effective at reducing casualties. Japan's drill culture saved many lives in 2011. But requires sustained investment and cannot protect against all scenarios.

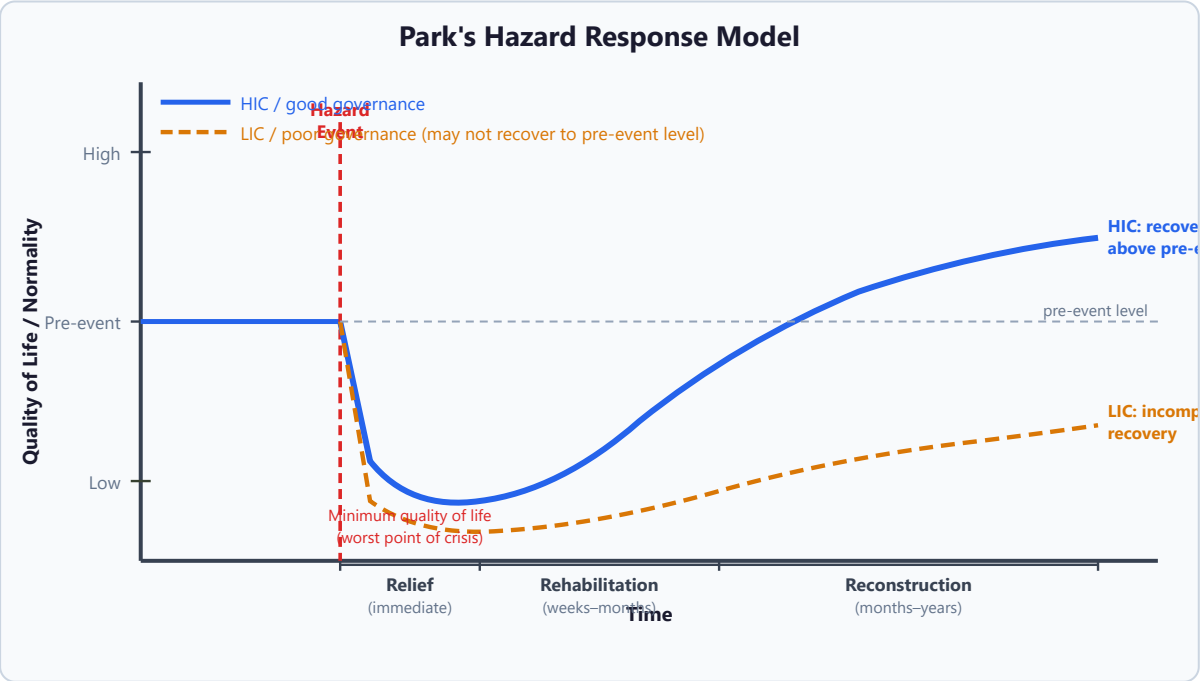
Adaptation	Living with hazards by modifying behaviour and land use	Land-use zoning (restricting building in high-risk areas), insurance schemes, diversified livelihoods, elevated buildings in flood zones	Effective long-term but politically difficult (people resist restrictions on their land). Insurance helps recovery but doesn't prevent damage.

The Hazard Management Cycle

Effective management is a continuous cycle: **Mitigation** (reducing risk before an event) → **Preparedness** (planning for response) → **Response** (immediate actions during/after event) → **Recovery** (rebuilding and learning lessons) → back to improved Mitigation. The cycle should improve with each iteration as lessons are learned. In practice, LICs often become trapped in the response-recovery phase without adequate investment in mitigation and preparedness.

Park's Hazard Response Model

Park's Model (1991) illustrates how a community's **quality of life** changes over time in response to a hazard event. It highlights the phases of response and the variable nature of recovery, particularly between HICs and LICs.



The Role of Governance and Development

The effectiveness of hazard management depends heavily on governance quality and level of development:

- **HICs** — generally have building codes, enforcement mechanisms, well-funded emergency services, early warning systems, insurance markets, scientific monitoring capacity, and political accountability. Deaths are lower but economic losses are higher in absolute terms.
- **LICs** — often lack building codes (or lack enforcement), have limited emergency services, weak monitoring networks, absent or inadequate insurance, rapid unplanned urbanisation in hazardous areas, and governance challenges. Deaths are much higher per event. 95% of disaster deaths occur in developing countries.
- **Fatalism vs adaptation** — cultural and religious attitudes influence how communities perceive and respond to hazard risk. Fatalism ("it's God's will") may reduce preparedness; adaptive cultures invest heavily in risk reduction.
- **Aid dependency** — frequent disasters can create dependency on international aid, undermining local capacity and governance. The "disaster-development nexus" means disasters set back development, while underdevelopment increases vulnerability — a vicious cycle.

Exam Tip: Edexcel 12-mark and 20-mark questions often require you to compare contrasting case studies (HIC vs LIC/LIDC). Always use paired comparisons with specific data. For example: "The 2011 Tohoku earthquake (M 9.0, Japan — HIC) killed 19,747 despite world-leading preparedness, primarily because the tsunami exceeded engineered defences. The 2010 Haiti earthquake (M 7.0 — LIC) killed 220,000+ despite being 500x less powerful, because extreme poverty, no building codes, and weak governance amplified vulnerability. This demonstrates that development level is often more important than hazard magnitude in determining impacts."

Exam Tip: For Edexcel, always consider the concept of **disaster risk** = hazard × vulnerability / capacity. An earthquake in an unpopulated desert is a hazard but not a disaster. The same earthquake in a densely populated city with weak buildings is a disaster. Vulnerability and capacity are as important as the physical hazard itself. This framework should underpin all your tectonic hazard answers.

Exam Tip: When discussing prediction and warning, be precise about what is and is not possible. Earthquakes CANNOT be reliably predicted (no proven method for forecasting timing, location, and

magnitude). Volcanic eruptions CAN be anticipated through precursory signs (increased seismicity, ground deformation, gas emissions, thermal changes) but false alarms and the exact timing of eruptions remain uncertain. Tsunamis CAN be detected after generation but warning time depends on distance from source. Do not claim earthquakes can be predicted in your exam answers.

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